

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1853

Jackson-Sunrise 100 kV
December 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1853. The Interconnection Customer proposes a 50.2 MW wind facility interconnecting at the Jackson-Sunrise 100 kV substation in the Illinois area. The request was received on August 10, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 580 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$906,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Jackson-Sunrise – A East Term 100kV
Contingency: Hill Country 138 230 kV/161 kV Auto
Loading: 105.7% (Rating: 146 MVA)
DFAX: 66.6%
Pre-project loading: 82.9%
2. Facility: Apison Pike – A E Term 100kV Line
Contingency: Hill Country 138 230-161 kV Transformer Bank
Loading: 100.3% (Rating: 140 MVA)
DFAX: 66.6%
Pre-project loading: 76.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jackson-Sunrise – A East Term 100kV	Reconductor line	\$1,170,000	30% (\$347,000)
A E Term – Apison Pike 100kV	Reconductor line	\$1,883,000	30% (\$559,000)
	Total allocated to project		\$906,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.